

On-Chain Digital Inheritance

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Introduction

Inheritance has served as a cornerstone of wealth transfer across civilizations for millennia, evolving through codified law and institutional frameworks designed for physical property and paper records. Yet these centuries-old mechanisms fail to serve most of the world’s population effectively: the majority of people globally die without wills, with procrastination and complexity cited as primary barriers rather than cost.

Cross-border inheritances face jurisdictional fragmentation, institutional intermediaries introduce trust dependencies and corruption risks, and updating arrangements requires expensive legal procedures that discourage timely revisions. The digital age makes this inheritance crisis acute. These universal barriers affect anyone attempting to plan inheritance, regardless of asset type or jurisdiction.

To understand the scope of this challenge, assets requiring inheritance fall into three distinct categories:

1. **Registry-bound assets where law mandates formal deeds** (real estate, titled vehicles, regulated securities)
2. **Institution-mediated assets without formal deed requirements** (online bank accounts, social media profiles, content channels, digital subscriptions, family keepsakes)
3. **Purely digital assets where access = ownership** (cryptocurrencies, encrypted files, private keys)

Traditional inheritance frameworks handle category 1 well, struggle with category 2, and fundamentally fail at category 3. Yet all three categories share common challenges: intermediary dependencies, trust requirements, inflexibility, and the inability to adapt to modern digital realities.

This white paper presents a solution that addresses these universal challenges through blockchain technology.

The Problem of Inheritance in the Digital Age

Traditional inheritance systems were designed centuries ago for physical property transfers within stable institutional frameworks. Yet these mechanisms fail to serve the majority of the world’s population effectively.

Accessibility Barriers: Studies show 50-70% of Americans die without wills, with similar patterns observed across Common Law jurisdictions—but the primary cause is not cost. Surveys reveal procrastination (“haven’t gotten around to it”) dominates at 43%, followed by complexity (“don’t know where to start”) at 30%. Traditional inheritance systems require confronting mortality, navigating complex legal procedures, coordinating with attorneys and notaries, and understanding specialized terminology—friction that causes indefinite postponement. The psychological and procedural hurdles—not just financial ones—prevent inheritance planning. In developing countries without established legal infrastructure, these barriers are even more acute, leaving assets in legal limbo and families without clear inheritance rights.

Trust Dependencies: Notaries and attorneys act as gatekeepers, introducing costs and potential corruption. Institutional intermediaries must remain solvent and trustworthy across decades—a precarious assumption in unstable jurisdictions. Cross-border inheritance faces jurisdictional complexity and institutional coordination challenges that often prove insurmountable.

Inflexibility: Updating Wills requires repeating expensive legal procedures. Life changes—new assets, additional beneficiaries, shifting circumstances—create friction that discourages timely updates. Disputes about legal capacity, document authenticity, or interpretation delay transfers for years.

Digital Age Disconnect: Systems designed for physical deeds and paper records struggle with digital credentials and encrypted assets. Traditional frameworks provide no mechanism for conditional access to information that updates continuously. Cryptographic keys cannot be safely held by intermediaries—if lost, stolen, or misused while in custody, assets become unrecoverable and professionals face liability, which is why most attorneys and notaries now refuse to handle digital inheritance.

These failures affect all three asset categories established in the Introduction—registry-bound assets, institution-mediated assets, and purely digital assets.

Yet inheritance introduces a unique challenge even for blockchain solutions: conditional transfer. Unlike immediate ownership transfers, inheritance requires assets to remain inaccessible to beneficiaries until verifiable conditions are met. This practical impasse stems from a deeper technical challenge rooted in the nature of cryptography itself.

The Cryptographic Paradox of Conditional Access

Tokenization enables digital representation of any asset on blockchain infrastructure, but conditional transfer—the core requirement for inheritance—presents a fundamental cryptographic paradox.

In traditional tokenization, possession equals capability. If a beneficiary holds the cryptographic keys to access a tokenized asset, they can access it immediately. Standard encryption does not inherently support temporal controls or conditional-release mechanisms. Once key material exists on-device or in a wallet, cryptographic security guarantees immediate access capability—delaying that access until future conditions are met becomes architecturally impossible through cryptographic means alone.

The challenge: how can a beneficiary possess a token representing an asset while being prevented from accessing that asset until verifiable conditions are met? In blockchain-based systems, this paradox is particularly acute. Public blockchains store all data transparently and permanently. If tokenized assets (encrypted representations) and decryption keys are both available to the beneficiary from the outset, no cryptographic mechanism can enforce conditional access—the beneficiary can activate the token (decrypt and access the assets) immediately, bypassing inheritance conditions entirely.

Many prior solutions have attempted to enforce control by withholding critical components (enforcing scarcity of key material), but in practice they introduce trust, coordination, or centralization assumptions that limit their robustness:

- **Multisig Collaborative Custody Models:** Solutions like Casa Covenant use multisignature architectures (typically 2-of-3 or 3-of-5) where users control the majority of keys while service providers hold recovery keys. Casa maintains true self-custody—their recovery key cannot unilaterally access funds and users retain full control. However, these approaches are human trust-bound, requiring coordination with estate lawyers or other designated keyholders. Additionally, the inheritance mechanism itself creates service availability dependencies: if the provider becomes unavailable, changes terms, or ceases operations, beneficiaries may face operational friction and potential access delays as the inheritance process depends on the provider’s participation and recovery key.
- **Secret Sharing Schemes:** Safe Haven Inheriti (Shamir’s Secret Sharing) and Vault12 (guardian networks) distribute control among multiple parties, relying on human coordination and social trust. They are human-trust-bound, requiring beneficiaries to coordinate with other keyholders, creating potential for disputes, coercion, or unavailability of required signatories.
- **Dead-man Switches:** Services like Sarcophagus use automated triggers with decentralized storage and economically-incentivized node operators who perform key management functions. While more decentralized than custodial approaches, these systems still introduce coordination dependencies and trust assumptions around operator behavior, despite economic incentive mechanisms. This design replaces institutional trust with incentive-aligned coordination among pseudonymous actors, but cannot fully eliminate behavioral assumptions or potential collusion among operators.
- **Time-lock Puzzles and VDFs:** Cryptographic mechanisms that create resource-bound delays based on computational work. While mathematically elegant, they are inherently inflexible—the delay is fixed at creation time and cannot adapt to external conditions like the testator’s actual status or changing circumstances.
- **DIY Solutions:** Hardware wallets in safes with seed phrases, passwords in sealed envelopes with lawyers, USB drives in safety deposit boxes, or encrypted files with keys split among family members. These approaches are fundamentally secrecy-bound and discovery-dependent, failing catastrophically when physical items are lost, damaged, stolen, or when technology becomes obsolete.

All these solutions reintroduce centralization, inflexibility, and trust dependencies that blockchain systems aim to eliminate.

Existing solutions face an inherent dilemma: they can either preserve decentralization by distributing all cryptographic materials to beneficiaries (enabling premature access), or enforce conditional access by withholding materials through trusted intermediaries (sacrificing decentralization).

No existing solution achieves both simultaneously.

Inheritor solves this paradox through a novel architectural breakthrough that makes tokenization with conditional access possible for the first time.

Inheritor’s Solution: The Architectural Time-Lock*

The cryptographic paradox of conditional access has been well documented in academic literature—including Frédéric Prost’s *Inheritance and Blockchain: Thoughts and Open Questions* (2022), Li et al.’s *Send Message to the Future? Blockchain-based Time Machines for Decentralized Reveal of Locked Information* (2024), Chen et al.’s *Beyond Life: A Digital Will Solution for Posthumous Data Management* (2025), and Yang and Yuan’s *Toward Timed-Release Encryption in Web3* (2022). These works consistently identify a fundamental tension: purely decentralized systems struggle to enforce conditional or event-based access without reintroducing external trust mechanisms such as oracles, governance votes, or key custodians.

Inheritor resolves this paradox through a novel architectural time-lock* mechanism, achieving conditional access through purely decentralized infrastructure.

Beneficiaries receive complete tokenized inheritance materials—both classical secp256k1 keys for blockchain operations and quantum-safe private keys (X-Wing) for asset decryption—from the moment the inheritance is created. All encrypted assets, metadata, and key encapsulation data are stored publicly on Arweave and Ethereum, fully accessible to the beneficiary. The token exists in the beneficiary’s possession from day one. Yet token activation is architecturally impossible. The enforcement emerges from binding key usability—not key possession—to blockchain-verified state.

On iOS, quantum-safe private keys are generated using platform APIs that prevent raw key extraction, with the root-of-trust anchored in Apple Silicon’s Secure Enclave. The Secure Enclave serves as the foundation of iOS keychain security: all keychain encryption keys are derived from the Secure Enclave’s hardware-unique identifier combined with the user’s passcode, and the Secure Enclave enforces code signing verification that restricts keychain access exclusively to authorized applications. X-Wing private keys are stored in iOS Keychain with iCloud synchronization enabled, allowing seamless cross-device access within the user’s Apple ecosystem while maintaining cryptographic binding to Apple’s platform through CryptoKit’s integrity-checked format.

Additional Authenticated Data (AAD) cryptographically binds each encrypted inheritance to its specific beneficiary’s public key via SHA-256 fingerprints in the key identifier, preventing key substitution attacks where an attacker with their own valid quantum-safe private key attempts to decrypt assets designated for other beneficiaries.

To ensure multi-generational accessibility regardless of platform evolution, each inheritance is encrypted for two quantum-safe recipients. The iOS-native path provides maximum transparency and verifiable enforcement through platform-level constraints, while the portable path ensures infrastructure independence through a lightweight conditional release service. Both paths enforce identical smart contract conditions and cryptographic binding. True decentralization emerges from this redundancy: the system doesn’t depend on iOS continuing to exist, the Inheritor app staying available, or the conditional release service remaining operational. Beneficiaries can use whichever path works decades into the future—no single platform, provider, or service can prevent access.

To demonstrate this infrastructure independence in practice, Inheritor has released open-source command-line tools into the public domain that enable anyone to access Digital Wills without the iOS app. While the smart contract requires signatures from the official Inheritor app to create new inheritances (preserving this as a commercial feature), community developers are free to build clients that read, verify, claim inheritances, and perform check-ins. The official Inheritor team maintains the reference implementation and stewardship of

the architectural time-lock mechanism, ensuring a single interoperable standard while welcoming community innovation to improve, extend, or integrate these tools into other projects.

The Inheritor application acts as gatekeeper, verifying smart contract state (Designated $\rightarrow$ Claimable $\rightarrow$ Claimed) before authorizing any decryption operation using these protected keys. Even if malicious code were introduced, the platform-level constraints prevent key exfiltration, and the publicly observable blockchain state ensures transparent verification of inheritance conditions.

This fundamentally differs from traditional key-withholding approaches: beneficiaries possess the complete token—all required cryptographic materials—yet the architecture ensures the token remains inert until verifiable on-chain conditions are satisfied. This is tokenization with a critical innovation: possession without premature activation.

This breakthrough creates tangible benefits for users:

- **Complete Transparency:** The system operates entirely through public infrastructure with no hidden intermediaries or private channels. Beneficiaries can independently verify that their inheritance setup uses only observable public blockchain infrastructure, eliminating dependencies on services that could fail, change terms, or require ongoing trust.
- **True Independence:** No trust dependencies of any kind are required. The inheritance operates autonomously through smart contracts and public blockchain infrastructure, freeing users from reliance on institutions (banks, attorneys, specialized services), human coordination (co-signers, guardians, verifiers), or incentivized operators (node networks, oracles). Unlike other solutions that merely shift trust from institutions to individuals or decentralized networks, Inheritor eliminates trust dependencies entirely through cryptographic enforcement and publicly verifiable blockchain state.
- **Decentralized:** Redundant access paths ensure inheritances remain accessible across decades, regardless of platform changes, corporate lifecycles, or service availability. No single provider can prevent access.
- **Verifiable Security:** Rather than asking users to trust promises, the architecture makes violations architecturally impossible or immediately detectable. For iOS users, the entire inheritance process operates through public infrastructure—network monitoring confirms communication only with Ethereum and Arweave, with no hidden API calls or secret data channels. Everything operates publicly except private keys, which iOS platform security prevents the app from accessing, even from the beneficiary.

The audit path is simple: observe network traffic to verify only public blockchain and storage connections with documented data formats. The architecture resists compromise even from malicious insiders or developers, providing security guarantees that can be independently verified rather than simply trusted.

This architectural breakthrough—conditional tokenization through the time-lock mechanism—enables On-Chain Digital Inheritance for the first time.

** Patent Pending*

Defining On-Chain Digital Inheritance

With the architectural time-lock solving the cryptographic paradox of conditional access, On-Chain Digital Inheritance becomes possible: blockchain tokenization applied to inheritance, creating digital representations of any asset with cryptographically enforced conditional transfer to designated beneficiaries.

Tokenization

On-Chain Digital Inheritance operates through pure blockchain tokenization. This differs markedly from “digital inheritance” services offering password vaults or digital wills—approaches that simply move centralized, trust-dependent models online. Instead, On-Chain Digital Inheritance leverages blockchain infrastructure: no institution holds your keys, no company needs to stay in business for decades, no recurring custody fees exist. Inheritance operates through decentralized networks with transparent, verifiable operations.

Each inheritance creates a unique, non-transferable token on-chain that represents conditional access rights to encrypted assets. These are soulbound conditional access tokens—a new category of specialized tokens specifically designed for inheritance. Each inheritance ID functions as a token with a unique identifier, assigned to a specific beneficiary, containing metadata (Arweave pointers, encrypted data), and tracking state changes through a state machine (Designated → Claimable → Claimed). Unlike standard NFTs that represent immediate ownership, inheritance tokens incorporate conditional activation where the token itself gates decryption rights. The architectural time-lock binds token usability to blockchain-verified conditions, creating tokenization with conditional access enforcement rather than simple asset representation.

Universal asset coverage

The tokenization model handles all three asset categories established in the Introduction identically through encryption, permanent storage, on-chain representation, and conditional access enforcement.

For native digital assets where possession constitutes ownership (cryptocurrencies, stablecoins, encrypted files, NFTs), users encrypt their private keys using quantum-safe encryption with beneficiaries designated from the outset. The encrypted materials become tokenized representations stored publicly on blockchain infrastructure where they cannot be lost or destroyed—eliminating the physical vulnerabilities of hardware wallets or paper seed phrases. iOS platform security prevents the app or even the beneficiary from extracting usable key material until smart contract conditions verify the token is claimable. This maintains the same ‘never expose your keys’ security standard while removing the risk of loss, theft, or premature access through discovery.

For registry-bound assets requiring formal documentation (real estate, vehicles, traditional securities), users work with legal professionals to prepare compliant transfer documents, which Inheritor then tokenizes and releases when smart contract conditions are met.

For institution-mediated assets (online accounts, credentials, personal instructions), Inheritor’s cryptographic proof of testamentary intent and asset documentation may be accepted by institutions as sufficient evidence of inheritance rights, potentially without requiring formal probate—though institutional policies vary.

What this means in practice

The tokenization approach combines cryptographic certainty—the math guarantees assets remain secure—with programmable flexibility through smart contracts that execute exact intentions. Everything operates transparently through public infrastructure that can be independently verified, giving users full control over their tokenized digital estate while ensuring beneficiaries have guaranteed access when the time comes, regardless of institutional changes, market conditions, or corporate lifecycles.

While the architectural time-lock mechanism has broad applicability across tokenization scenarios—from time-locked release of confidential information and corporate succession planning to whistleblower dead-man switches, escrow arrangements, and conditional access to medical or legal records—Inheritor specifically deploys this breakthrough to solve the inheritance challenge.

Where other conditional tokenization applications may emerge over time, the need for secure, trustless On-Chain Digital Inheritance is immediate and universal, affecting anyone who holds assets and wants to ensure their beneficiaries can access them.

Building a Solution for the Ages

Building a practical tokenization system for conditional inheritance requires careful attention to blockchain infrastructure, permanent storage, condition verification, user experience, and multi-generational resilience. Like all blockchain tokenization, Inheritor creates digital representations on-chain, but the architectural time-lock introduces novel enforcement mechanisms that require thoughtful implementation. Inheritor addresses each component systematically.

Flexible Blockchain Strategy: Security or Cost Efficiency

Inheritor offers users a choice between Ethereum Layer 1 and Arbitrum for each inheritance they create. Ethereum L1 provides maximum security and decentralization—ideal for high-value assets (cryptocurrency wallets, property deeds) or inheritances where permanence on the most secure layer matters regardless of monetary value (legal documents, irreplaceable records). Arbitrum reduces fees by over 90% through Optimistic Rollups, making it suitable for lower-stakes inheritances (family photos, password lists, social media accounts) while still leveraging Ethereum’s security through fraud proofs and 7-day dispute windows. Users can mix and match blockchains across their inheritances, optimizing each transfer based on the specific asset’s value, nature, and criticality. (Note: Cardano integration is on our roadmap.)

Permanent Decentralized Storage with Quantum-Safe Encryption

Standard blockchain tokenization typically stores minimal metadata on-chain (contract state, token IDs) with larger assets referenced via IPFS or centralized storage. However, inheritance requires permanent, immutable storage that persists across generations without recurring costs or maintenance. Inheritor leverages Arweave for decentralized, permanent storage of tokenized assets. Arweave’s “pay once, store forever” model eliminates recurring fees while ensuring data permanence through its append-only Blockweave architecture. All tokenized assets are encrypted using X-Wing quantum-safe cryptography, protecting inheritances against future quantum computing threats across generations. This combination creates true multi-generational security: permanent storage with cryptographic protection that remains effective regardless of technological advances.

Asset Tokenization: Universal File Upload

Inheritor’s tokenization approach is deliberately universal and format-agnostic. Users can upload any digital file—text documents, PDFs, photos, videos, spreadsheets, or any other file format—to create a tokenized inheritance. This flexibility allows users to define what constitutes “the inheritance” based on their specific needs: encrypted cryptocurrency wallet files and recovery phrases, pre-signed legal documents prepared with attorneys, scanned property deeds, login credentials for digital accounts, personal instructions for beneficiaries, family photos and videos with sentimental value, or any combination thereof. Each inheritance is simply an encrypted file stored permanently on Arweave, with no restrictions on content or format.

To provide context without compromising security, testators can optionally include a short message that remains encrypted but becomes visible to beneficiaries when the inheritance is designated. This message helps beneficiaries understand what an inheritance contains (“Your grandmother’s photo collection from 1960-1985” or “Login credentials for family Netflix account”) without revealing the actual encrypted contents.

Inheritor also supports anonymous inheritance creation. When enabled, beneficiaries see only that an inheritance has been designated for them, without any indication of the testator’s identity in the user interface. The blockchain transaction remains publicly verifiable, but the application layer provides privacy. This feature serves various use cases: surprise inheritances from distant relatives, privacy-conscious individuals who prefer discretion, or complex family situations where testator anonymity may be appropriate. The beneficiary learns the testator’s identity only when they decrypt the inherited assets, if the testator chose to include that information within the encrypted contents.

Condition Verification: Check-Ins with Optional Human Oversight

The architectural time-lock enforces conditional access through smart contract verification—but how does the system determine when conditions are met without depending on centralized death verification services or institutional attestors? Inheritor implements a check-in mechanism with optional human verification.

Testators confirm their wellbeing through periodic check-ins—a simple interaction taking less than 10 seconds. Missed check-ins trigger a grace period that is independently configurable for each inheritance, enabling precise control over asset release timing. One inheritance might expire 30 days after the final check-in, releasing immediate necessities to beneficiaries, while another might remain locked for a year or longer, deferring access to certain assets until beneficiaries are ready to manage them. When each inheritance’s grace period expires, the smart contract transitions to claimable state upon the next user interaction—typically when beneficiaries check inheritance status or attempt to claim. This user-triggered evaluation eliminates dependency on external automation services (like Chainlink Keepers) or death verification oracles for condition verification, as

beneficiaries have natural economic incentive to verify claimability themselves. This per-inheritance timing operates entirely through blockchain interaction, requiring no external verification infrastructure.

For additional protection against accidental expiration, testators can optionally designate a beneficiary as a verifier. A verifier is a beneficiary who receives their own designated inheritance and must confirm the testator's condition on behalf of all beneficiaries before any inheritances become claimable. When check-ins are missed and verification is enabled, verifiers assess whether the testator is truly incapacitated or merely traveling, ill, or temporarily unavailable, then submit their attestation as an on-chain transaction. Upon confirmation of death, all inheritances become claimable after their respective grace periods expire—including the verifier's own inheritance. This design involves an explicit trade-off: human judgment provides protection against premature claiming due to technical issues or temporary unavailability, but introduces a trust assumption that verifiers will act honestly. Testators choose this trade-off themselves—opting for either purely automated, trustless execution based solely on check-in timing, or adding a human verification layer that introduces trust in exchange for additional safeguards. Verifier status can be assigned, enabled, or disabled at any time through blockchain transactions.

Seamless User Experience Through Account Abstraction

Inheritor eliminates blockchain complexity through ERC-4337 account abstraction. Gas fees are sponsored by a paymaster and recovered through in-app purchases—creating a gasless user experience where users never need to hold ETH. On-chain actions feel like standard iOS interactions while maintaining verifiable execution. Importantly, the underlying smart contracts remain accessible via traditional externally owned accounts, ensuring infrastructure independence if the app or paymaster becomes unavailable.

Together, these implementation choices—flexible blockchain deployment, permanent quantum-safe storage, automated condition verification, and abstracted user interaction—create a tokenization system that is secure, affordable, easy to use, and accessible across generations.

Conclusion

Blockchain tokenization has transformed how we represent and transfer value, but conditional transfers—where tokens activate only when verifiable conditions are met—have remained elusive without trusted intermediaries. For the first time, Inheritor's architectural time-lock* mechanism resolves this fundamental paradox, enabling truly trustless conditional tokenization.

The breakthrough lies in separating token possession from token activation. Beneficiaries receive complete tokenized inheritance materials from day one—all cryptographic keys, all encrypted assets, all metadata stored publicly on blockchain infrastructure. Yet the architecture ensures these materials remain inert until verifiable on-chain conditions are satisfied.

The implications extend across asset classes. Cryptocurrency holders can tokenize their wallets for inheritance with greater safety than hardware wallets while guaranteeing beneficiary access. Families can tokenize login credentials, encrypted files, and personal instructions without trusting third parties. Legal documents and property deeds can be tokenized and released automatically when conditions are met, integrating with—rather than replacing—traditional legal frameworks.

As blockchain tokenization grows toward McKinsey's projected \$2 trillion market by 2030, conditional transfers will become increasingly critical. Inheritor's architectural time-lock* provides the missing infrastructure for this evolution—enabling tokenized assets to transfer according to programmable conditions without reintroducing centralization. Inheritances created today will execute decades into the future through public blockchain infrastructure that anyone can verify and no one can shut down.

Beyond the technical innovation, Inheritor democratizes inheritance planning. Creating a Digital Will—documenting your testamentary intent and securing your complete digital estate for designated loved ones—becomes as simple as using your phone from home. Every asset can be secured for those you leave behind—not as a distant obligation requiring ongoing professional assistance, but as an expression of responsibility and love, maintained with the same ease as managing any other aspect of your digital life.

** Patent Pending*

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Disclaimer

This white paper is for informational purposes only and does not constitute legal, financial, or investment advice. Users are responsible for ensuring their inheritance instructions comply with applicable laws including forced heirship requirements, tax obligations, and jurisdictional regulations.

Inheritor does not create legally valid wills for probate purposes in the traditional sense. Users with significant registry-bound assets should consult estate planning attorneys to ensure compliance with jurisdictional will execution requirements. Consult with professional advisors before making any inheritance decisions.